

Synthesis of Polypropylene Graft Copolymers by the Combination of a Polypropylene Copolymer Containing Pendant Vinylbenzene Groups and Atom Transfer Radical Polymerization

CHENGANG CAO,^{1,2} JUNFENG ZOU,^{1,2} JIN-YONG DONG,¹ YOU LIANG HU,¹ T. C. CHUNG³

¹CAS Key Laboratory of Engineering Plastics, Joint Laboratory of Polymer Science and Materials, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100080, China

²Graduate School, Chinese Academy of Sciences, Beijing 100080, China

³Department of Materials Science and Engineering, Pennsylvania State University, University Park, Pennsylvania 16802

Received 26 May 2004; accepted 3 September 2004

DOI: 10.1002/pola.20509

Published online in Wiley InterScience (www.interscience.wiley.com).

ABSTRACT: This article discusses a facile and inexpensive reaction process for preparing polypropylene-based graft copolymers containing an isotactic polypropylene (*i*-PP) main chain and several functional polymer side chains. The chemistry involves an *i*-PP polymer precursor containing several pendant vinylbenzene groups, which is prepared through the Ziegler–Natta copolymerization of propylene and 1,4-divinylbenzene mediated by an isospecific MgCl₂-supported TiCl₄ catalyst. The selective monoenchainment of 1,4-divinylbenzene comonomers results in pendant vinylbenzene groups quantitatively transformed into benzyl halides by hydrochlorination. In the presence of CuCl/pentamethyldiethylenetriamine, the *in situ* formed, multifunctional, polymeric atom transfer radical polymerization initiators carry out graft-from polymerization through controlled radical polymerization. Some *i*-PP-based graft copolymers, including poly(propylene-*g*-methyl methacrylate) and poly(propylene-*g*-styrene), have been prepared with controlled compositions. © 2004 Wiley Periodicals, Inc. *J Polym Sci Part A: Polym Chem* 43: 429–437, 2005

Keywords: atom transfer radical polymerization (ATRP); divinylbenzene; graft copolymers; isotactic; poly(propylene) (PP); vinylbenzene groups

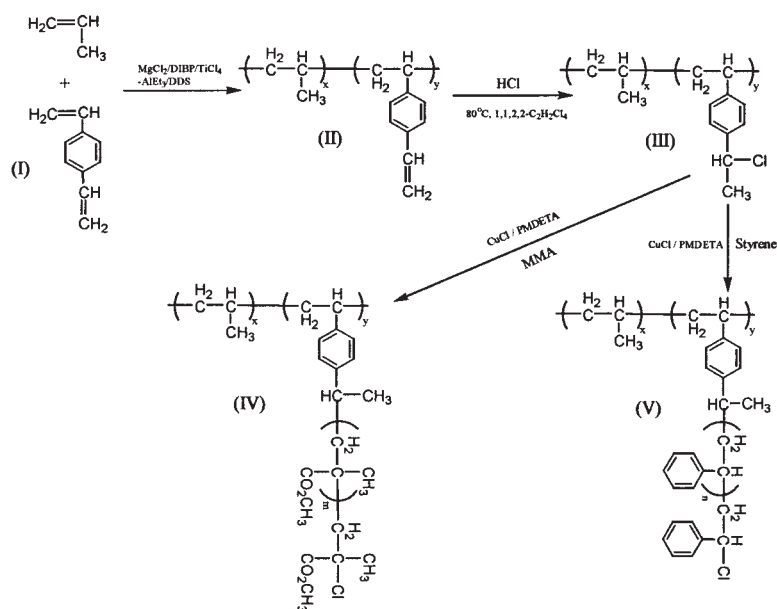
INTRODUCTION

Polypropylene (PP) is one of the most important polymeric materials.^{1,2} It has excellent physical mechanical properties and thus a wide range of applications.^{3,4} However, because of the lack of polar functional groups on its polymer chain, PP

usually suffers from poor adhesion to and incompatibility with other polymers or materials, and this has significantly limited many of its end uses. To resolve this problem, one established approach is to introduce a second functional polymeric component as a side chain to synthesize PP graft copolymers that not only provide a large quantity of functional groups but also preserve the original desired PP properties, such as the crystallinity (χ_c), melting temperature (T_m), and hydrophobicity.⁵

Correspondence to: J.-Y. Dong (E-mail: jydong@iccas.ac.cn)

Journal of Polymer Science: Part A: Polymer Chemistry, Vol. 43, 429–437 (2005)
© 2004 Wiley Periodicals, Inc.



Scheme 1

Atom transfer radical polymerization (ATRP) is a recently developed quasiling polymerization technique that can polymerize a variety of polar and nonpolar vinyl monomers in a controlled manner.^{6–10} The controlled chain growth, in combination with the living nature, makes ATRP a very valuable tool for engineering macromolecular architectures, including block and graft copolymers. With the grafting-from or macromonomer technique, graft copolymers of various backbones and grafts have been realized.^{11–18} Among them, polyolefin-based graft copolymers are particularly interesting, in that they are effective compatibilizers in blends of polyolefins with other polymeric materials. For this reason, much effort has lately been paid to the synthesis of polyolefin graft copolymers via ATRP.^{19–28} Matyjaszewski et al.¹⁹ first reported the conversion of commercially available poly(ethylene-*co*-glycidyl methacrylate) into ATRP macroinitiators by a reaction with chloroacetic acid or 2-bromoisobutyric acid and the subsequent preparation of poly(ethylene-*g*-styrene) and poly(ethylene-*g*-methyl methacrylate) graft copolymers with well-controlled graft structures. Later, several effective methods were laid out for the ATRP synthesis of polyethylene (PE)-based graft copolymers. Some major examples include (1) the selective bromination of metallocene-catalyzed ethylene–styrene random copolymers at the benzylic positions and the synthesis of several PE graft and graft–block copol-

ymers,²⁰ (2) the ATRP copolymerization of *n*-butyl acrylate with a methacrylate-terminated branched PE macromonomer synthesized from Pd-mediated living ethylene polymerization,²¹ and (3) the reverse ATRP of methyl methacrylate (MMA) onto γ -ray-modified peroxide-containing PE.²² However, in contrast to the relatively versatile approaches for PE graft copolymers, there are only a few reports discussing the ATRP synthesis of graft copolymers based on isotactic polypropylene (*i*-PP), including poly(propylene-*g*-styrene) prepared via the metallocene-mediated copolymerization of propylene with an ATRP-obtained allyl-terminated polystyrene (PS) macromonomer²³ and photobrominated *i*-PP thin films grafted with *N*-isopropyl acrylamide.²⁴

In this article, we present a convenient yet highly efficient method for the preparation of *i*-PP-based graft copolymers via ATRP, as illustrated in Scheme 1. In this method, a small number of vinylbenzene groups are first added to *i*-PP chains by the isospecific Ziegler–Natta-catalyzed copolymerization of propylene and 1,4-divinylbenzene through monoenchainment. The obtained *i*-PP containing pendant vinylbenzene groups then reacts with dry hydrogen chloride (HCl) to form polymeric benzyl halide initiator sites, which, in the presence of CuCl/pentamethyldiethylenetriamine (PMDTA), allow for the ATRP synthesis of *i*-PP-based graft copolymers, including poly(propylene-*g*-methyl methacrylate) (*i*-PP-

g-PMMA) and poly(propylene-*g*-styrene) (*i*-PP-*g*-PS).

EXPERIMENTAL

Materials

All O₂- and moisture-sensitive manipulations were carried out inside an argon-filled, vacuum-atmosphere drybox equipped with a dry train. Pure 1,4-Divinylbenzene was isolated from a commercially available divinylbenzene mixture (Aldrich Chemical Co.) according to a literature procedure.²⁹ Styrene (99%), and MMA (99%) were vacuum-distilled and stored under argon. Anisole (99%) was deoxygenated by an argon purge before being refluxed for 24 h and was then distilled over CaH₂. Heptane was deoxygenated by an argon purge before being refluxed for 48 h and then was distilled over sodium. AlEt₃ (1.8 M in toluene), dimethoxydiphenylsilane (DDS), CuCl (98%), and PMDETA (99%) were acquired from Aldrich and used without further purification. Polymerization-grade propylene was supplied by Yansan Petrochemical Co. (China). The MgCl₂/diisobutyl phthalate (DIBP)/TiCl₄ catalyst was kindly donated by Xiangyang Chemical Co. (China) and had a Ti concentration of 2.5 wt %.

Characterization

All ¹H NMR spectra were recorded on a Bruker AM-300 instrument in 1,1,2,2-tetrachloroethane-*d*₂ at 110 °C. The *T_m*'s, glass-transition temperatures (*T_g*'s), and χ_c 's of the polymers were measured by differential scanning calorimetry (DSC) with a PerkinElmer DSC-7 instrument controller operating from 30 to 180 °C at a heating rate of 20 °C/min. The data were collected during the second heating cycle. The intrinsic viscosity ($[\eta]$) of the polymers was measured in decahydronaphthalene at 135 ± 0.1 °C. High-temperature gel permeation chromatography (GPC) measurements were performed with a Waters Alliance GPC 2000V instrument operated at 150 °C with 1,2,4-trichlorobenzene as the eluent.

Synthesis of *i*-PP Containing Pendant Vinylbenzene Groups

Heptane (50 mL) was added to a 250-mL, three-necked glass reactor equipped with a magnetic stirrer. The reactor was then saturated with at-

mospheric-pressure propylene at 50 °C. 1,4-Divinylbenzene (1 g), 3.88 mL of AlEt₃ (1.8 M in toluene), and 4.13 mL of DDS (0.7 mmol) were subsequently injected with a syringe with an Al/Si molar ratio of 10/1. The polymerization was started by the addition of 0.1 g of the MgCl₂/DIBP/TiCl₄ catalyst to the reactor. After 30 min of reaction, 10 mL of acidified methanol was added to terminate the reaction. The product was collected by filtration, and this was followed by repeated washing with acidified ethanol, tetrahydrofuran (THF), and water to remove the catalyst residue and the unreacted 1,4-divinylbenzene comonomer. The polymer was then dried at 60 °C under a high vacuum for 24 h to obtain 5.36 g of a white polymer powder.

Hydrochlorination of *i*-PP Containing Pendant Vinylbenzene Groups

The hydrochlorination reaction was carried out in a 250-mL, round-bottom flask equipped with a magnetic stirrer. Before the reaction, the raw material, the *i*-PP polymer containing pendant vinylbenzene groups, was purified once again by dissolution in boiling xylene followed by precipitation in methanol to ensure the complete removal of any inorganic impurities. For the hydrochlorination reaction, 2.0 g of styrene-pending *i*-PP polymer was suspended in 100 mL of 1,1,2,2-tetrachloroethane at 80 °C. Dry HCl, which was generated by the dropwise addition of HCl (36%) to concentrated sulfuric acid, was bubbled into the suspension. After 6 h, the suspension was poured into 200 mL of ethanol, filtered, and washed with ethanol several times. The final product was dried at room temperature under a high vacuum for 24 h and was obtained as a white polymer powder [i.e., *i*-PP polymer containing pendant 1-chloroethylbenzene groups (*i*-PP-Bz-Cl)].

Synthesis of *i*-PP-*g*-PMMA

A dry, round-bottom flask was charged with anisole (10 mL), CuCl (0.043 g, 0.3 mmol), PMDETA (0.051 g, 0.3 mmol), MMA (3.0 g, 30.0 mmol), and *i*-PP-Bz-Cl (0.15 g). The flask was sealed and filled with purified argon. After the mixture was allowed to stir at the ambient temperature for 5 min, the flask was placed in an oil bath at 95 °C for 1.5 h. The reaction was terminated by the contents of the flask being poured into a large amount of acidic methanol. The precipitated poly-

mer was washed and dried *in vacuo*. The obtained polymer was slightly yellow because of metal contamination. To make it colorless, the polymer was first dissolved in boiling xylene and was then precipitated in a large amount of methanol.

Synthesis of *i*-PP-*g*-PS

A dry round-bottom flask was charged with anisole (10 mL), CuCl (0.3 mmol), PMDETA (0.051 g, 0.3 mmol), styrene (3.0 g, 29.0 mmol), and *i*-PP-Bz-Cl (0.15 g). The flask was sealed and filled with purified argon. After the mixture was allowed to stir at the ambient temperature for 5 min, the flask was placed in an oil bath at 125 °C for 1.5 h. The reaction was terminated by the contents of the flask being poured into a large amount of acidic methanol. The precipitated polymer was washed and dried *in vacuo*. The obtained polymer was slightly yellow because of metal contamination. To make it colorless, the polymer was first dissolved in boiling xylene and was then precipitated in a large amount of methanol.

RESULTS AND DISCUSSION

Synthesis and Characterization of the *i*-PP Containing Pendant Vinylbenzene Groups

To demonstrate our facile chemistry for the ATRP synthesis of *i*-PP-based graft copolymers, we synthesized an *i*-PP polymer containing pendant vinylbenzene groups through the copolymerization of atmospheric-pressure propylene with 0.154 mol/L of 1,4-divinylbenzene catalyzed by an isospecific MgCl₂-supported Ziegler–Natta catalyst in 50 mL of anhydrous heptane at 50 °C. The catalyst activity of the copolymerization reaction was 180 kg of polymer/mol of Ti h, which was slightly lower than that of the propylene homopolymerization (236 kg of polymer/mol of Ti h under similar conditions). The ¹H NMR analysis of the polymer showed that about 0.53 mol % of the 1,4-divinylbenzene units were incorporated into PP chains with a conversion of 8.8%, and the 1,4-divinylbenzene incorporation proceeded via a selective mono-*enchainment* mode. Figure 1(a) shows the ¹H NMR spectrum of the polymer. In addition to the major peaks at 0.95, 1.35, and 1.65 ppm corresponding to the CH₃, CH₂, and CH groups in the propylene units, several minor chemical shifts can be observed at 5.2 and 5.7 ppm (doublet, CH=CH₂) and at 6.7 ppm (dou-

blet of doublet, CH=CH₂), and aromatic proton peaks can be observed at 7.1 and 7.4 ppm (C₆H₄). The integrated intensity ratio for the peaks at 5.2, 5.7, 6.7, 7.1, and 7.4 ppm is about 1:1:1:2: 2, and this suggests that no double *enchainment* of 1,4-divinylbenzene occurred during the polymerization and that the polymer obtained possessed a linear structure, which must be attributed to the combined advantages of the use of the Ziegler–Natta catalyst, which showed a poor capacity for incorporating bulky monomers, and the enhanced steric hindrance by the polymer chain at the residual double bond of the incorporated 1,4-divinylbenzene unit, which effectively prevented the residual double bonds from further participation in the polymerization reaction. This mechanism is analogous to that of a specific metallocene catalyst being used to achieve a linear copolymerization of ethylene and 1,4-divinylbenzene,²⁹ only with much reduced 1,4-divinylbenzene incorporation. The polymer is completely soluble in xylene, tetrachloroethane, decahydronaphthalene, and 1,2,4-trichlorobenzene at elevated temperatures. The [η] measurement of the polymer gave a value of 1.50 dL/g, which accounted for a viscosity-average molecular weight (*M*_η) of about 1.47 × 10⁵ g/mol.³⁰ Its GPC analysis revealed a number-average molecular weight (*M*_n) and molecular weight distribution of 3.7 × 10⁴ g/mol and 7.1, respectively. Given that the incorporation of 1,4-divinylbenzene in the polymer was 0.53 mol %, we roughly estimated the amount of 1,4-divinylbenzene inserted into each polymer chain to be 4.67 unit/chain. Because of its relatively low 1,4-divinylbenzene incorporation, the polymer preserved most of the desired properties of *i*-PP. It had a *T*_m value of 156.7 °C and a χ_c value of 58.8%, which were measured by DSC (Fig. 2).

Transformation of the Pendant Vinylbenzene Groups into ATRP Initiator Sites

The hydrochlorination of the *i*-PP polymer containing pendant vinylbenzene groups was carried out via the bubbling of dry HCl into a flask containing a suspension of the powdery polymer in 1,1,2,2-tetrachloroethane at 80 °C for 6 h. The hydrochlorinated product was also completely soluble in tetrachloroethane or decahydronaphthalene at elevated temperatures. Its ¹H NMR spectrum is shown in Figure 1(b). In comparison with the ¹H NMR spectrum of the starting *i*-PP containing pendant vinylbenzene groups, all the peaks corresponding to the vinyl group com-

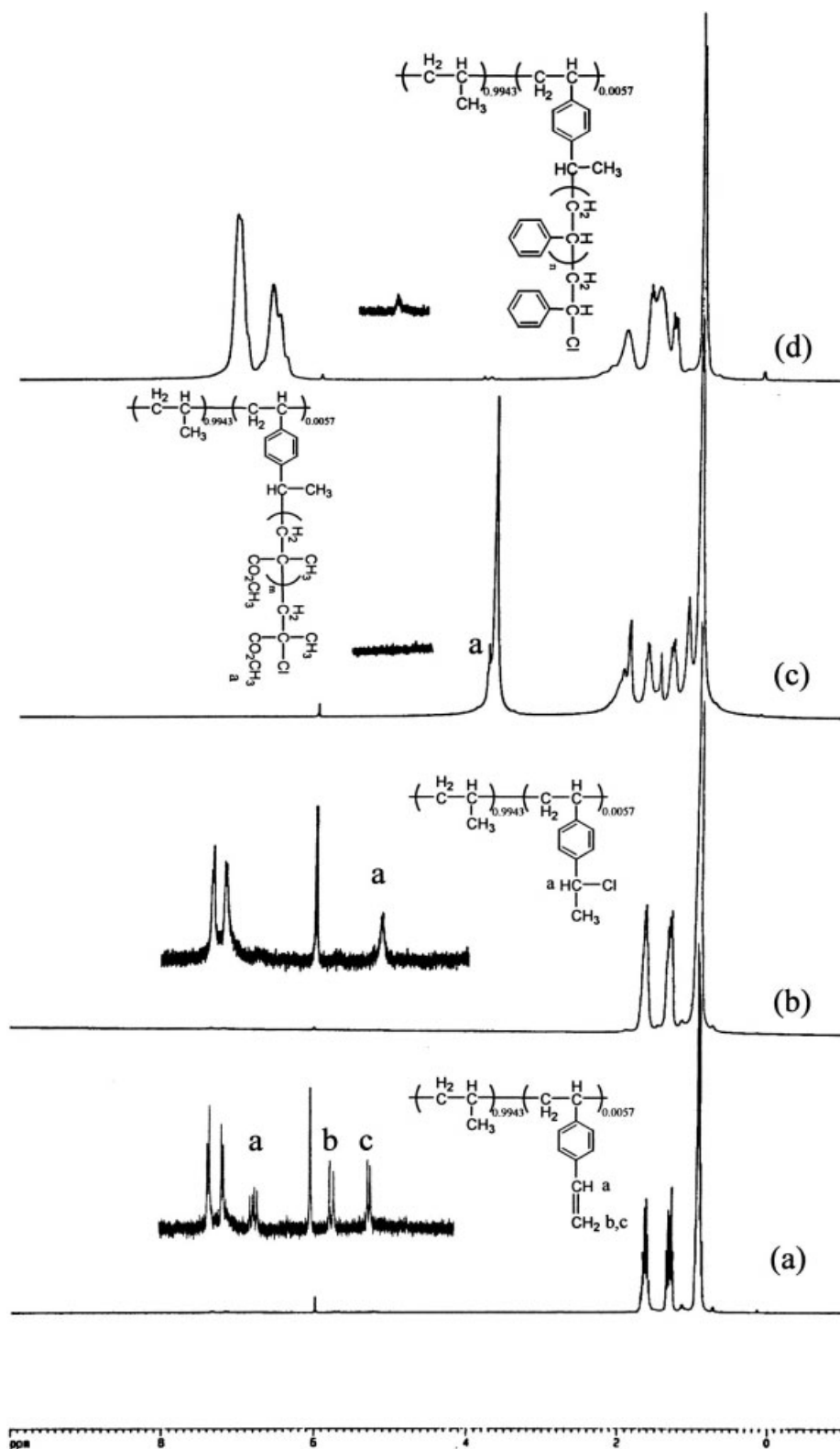


Figure 1. ^1H NMR spectra of (a) the poly(propylene-co-1,4-divinylbenzene) copolymer containing 0.53 mol % pendant vinylbenzene groups, (b) the hydrochlorinated poly(propylene-co-1,4-divinylbenzene) copolymer, (c) the *i*-PP-*g*-PMMA graft copolymer containing 56.4 mol % MMA units, and (d) the *i*-PP-*g*-PS graft copolymer containing 64.2 mol % styrene units (solvent = 1,1,2,2-tetrachloroethane- d_2 ; temperature = 110 $^\circ\text{C}$).

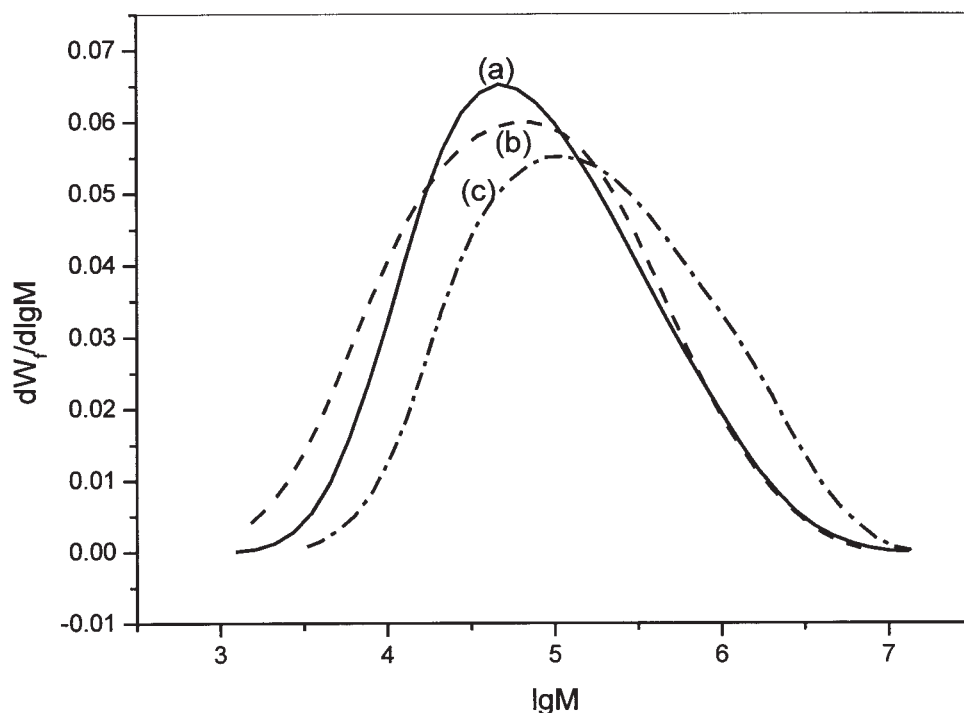


Figure 2. GPC profiles of (a) the starting hydrochlorinated poly(propylene-*co*-1,4-divinylbenzene) copolymer, (b) the graft copolymer *i*-PP-*g*-PMMA, and (c) the graft copolymer *i*-PP-*g*-PS.

pletely disappear after hydrochlorination. However, a new peak corresponding to the benzyl chloride proton appears at 5.1 ppm. The integration ratio of the peaks of the benzyl chloride proton and the aromatic protons is about 1/4, indicating that, despite the heterogeneous reaction conditions, the transformation of the pendant vinylbenzene groups into 1-chloroethylbenzene groups is quantitative. Viscosity measurements performed on the hydrochlorinated polymer showed no change with respect to its starting counterpart.

Synthesis of the *i*-PP-Based Graft Copolymers

i-PP-Bz-Cl is an effective polymeric, multifunctional ATRP initiator. *i*-PP-based graft copolymers with well-defined structures, that is, defined graft density and graft length, are accessible by ATRP graft polymerization from *i*-PP-Bz-Cl.

In two typical examples, *i*-PP-Bz-Cl was used to initiate the graft polymerization of MMA and styrene in the presence of the CuCl/PMDETA catalyst. For these two graft polymerizations, the molar ratio of CuCl, PMDETA, and the initiating C—Cl group was set at 3/3/1. The graft polymer-

ization reactions were conducted in anisole at 95 (for MMA polymerization) and 125 °C (for styrene polymerization). Even though the *i*-PP-Bz-Cl polymers were not completely soluble under the adopted reaction conditions, the graft reactions were still very effective. *i*-PP-*g*-PMMA containing 56.4 mol % MMA units and *i*-PP-*g*-PS containing 64.2 mol % styrene units were successfully obtained in 1.5-h reactions. Their ¹H NMR spectra are shown in Figure 1(c,d). In addition to the major chemical shifts at 0.95, 1.35, and 1.65 ppm corresponding to the protons in the backbone propylene units, the two spectra exhibit strong resonant signals representing the corresponding grafting segments. For the *i*-PP-*g*-PMMA graft copolymer, the strong peak observed at 3.68 ppm is the characteristic resonance of —COOCH₃ in MMA units. For the *i*-PP-*g*-PS graft copolymer, the spectrum shows the chemical shifts of the protons on the styrene repeating units in the PS grafting segments at 7.1, 6.5, 2.2, and 1.5 ppm.

To prove and define the graft structures of the obtained *i*-PP-*g*-PMMA and *i*-PP-*g*-PS graft copolymers, we studied the details of their ¹H NMR spectra, along with the results of the GPC characterization and the selective solvent extraction

Table 1. GPC Results for the Molecular Weights and Molecular Weight Distributions of the Poly(propylene-co-1,4-divinylbenzene) Copolymer, its Hydrochlorinated Product, and the Resultant *i*-PP-*g*-PMMA and *i*-PP-*g*-PS Graft Copolymers

Polymer	M_n ($\times 10^4$ g/mol)	PDI (M_w/M_n) ^a
Poly(propylene-co-1,4-divinylbenzene)	3.7	7.1
Hydrochlorinated poly(propylene-co-1,4-divinylbenzene)	3.7	7.2
<i>i</i> -PP- <i>g</i> -PMMA	4.8	7.4
<i>i</i> -PP- <i>g</i> -PS	5.3	7.9

^a Polydispersity index (i.e., molecular weight distribution).

performed on them. In the ^1H NMR spectrum of the *i*-PP-*g*-PMMA copolymer, the aforementioned resonance at 5.1 ppm for the starting *i*-PP-Bz-Cl polymer representing the benzyl chloride proton is no longer observable. Instead, a new, minor peak that is attributed to $-\text{COOCH}_3$ groups in the terminal halide-capped MMA units appears at 3.82 ppm, and it indicates that the poly(methyl methacrylate) (PMMA) grafting chains did grow from *i*-PP-Bz-Cl. In fact, a control experiment of MMA polymerization conducted in the absence of *i*-PP-Bz-Cl under otherwise identical ATRP reaction conditions yielded no polymer product. Boiling acetone extraction of the *i*-PP-*g*-PMMA graft copolymer resulted in negligible amounts of extractable substances. The combined experimental observations adequately prove the exclusive formation of the *i*-PP-*g*-PMMA graft copolymer with no contamination with ungrafted PMMA homopolymer. For the *i*-PP-*g*-PS graft copolymer, similar experiments aiming to confirm its graft structure were also performed, including a control experiment of styrene polymerization in the absence of *i*-PP-Bz-Cl and a boiling THF extraction of the graft copolymer. Both experiments came out with positive results. However, in the ^1H NMR spectrum of the graft copolymer, a very weak resonant signal of the benzyl chloride proton was observed, and it implied that not all of the initiating sites of C—Cl groups in *i*-PP-Bz-Cl were converted into living radical centers during the ATRP graft polymerization of styrene. The conversion efficiency of the potential initiating sites to its true counterparts is definitely related to the reaction conditions, which will certainly need to be optimized for the best grafting reaction. However, at this stage, we would rather quantitate the initiation efficiency of the current ATRP graft polymerization. By comparing the ra-

tios of the integrations of the peaks at 5.1 and 0.95 ppm for the starting *i*-PP-Bz-Cl polymer and the resulting *i*-PP-*g*-PS graft copolymer, we obtained a value of 82%.

The GPC profiles of the two graft copolymers and the starting *i*-PP-Bz-Cl polymer are shown in Figure 2. The data for the molecular weights and molecular weight distributions of the polymers are listed in Table 1. The M_n and molecular weight distribution values of the starting *i*-PP-Bz-Cl polymer were calculated (by comparison with narrowly distributed PS standards) to be 3.7×10^4 g/mol and 7.2, respectively. Neither the *i*-PP-*g*-PMMA graft copolymer nor the *i*-PP-*g*-PS graft copolymer showed significantly broadened molecular weight distributions (7.40 for *i*-PP-*g*-PMMA and 7.95 for *i*-PP-*g*-PS) in comparison with that of the starting *i*-PP-Bz-Cl. The GPC profiles of both graft copolymers show single peaks, further proving the exclusive formation of graft copolymers by the ATRP graft polymerization process.

As the graft structures of the two graft copolymers were proved, the parameters for the graft structure, namely, the graft density and graft length, could be defined. For the *i*-PP-*g*-PMMA graft copolymer, because the resonance at 5.1 ppm in the ^1H NMR spectrum of *i*-PP-Bz-Cl representing the benzyl chloride proton was no longer observable in the spectrum of the graft copolymer, the initiation efficiency was estimated to be 100%. Therefore, the graft density of the graft copolymer could be considered to be equal to the concentration of 1-chloroethylbenzene groups in *i*-PP-Bz-Cl. As 1-chloroethylbenzene groups were transformed from vinylbenzene groups also in a quantitative manner, the graft density could be further reckoned by the density of the pending vinylbenzene groups in the original poly(propyl-

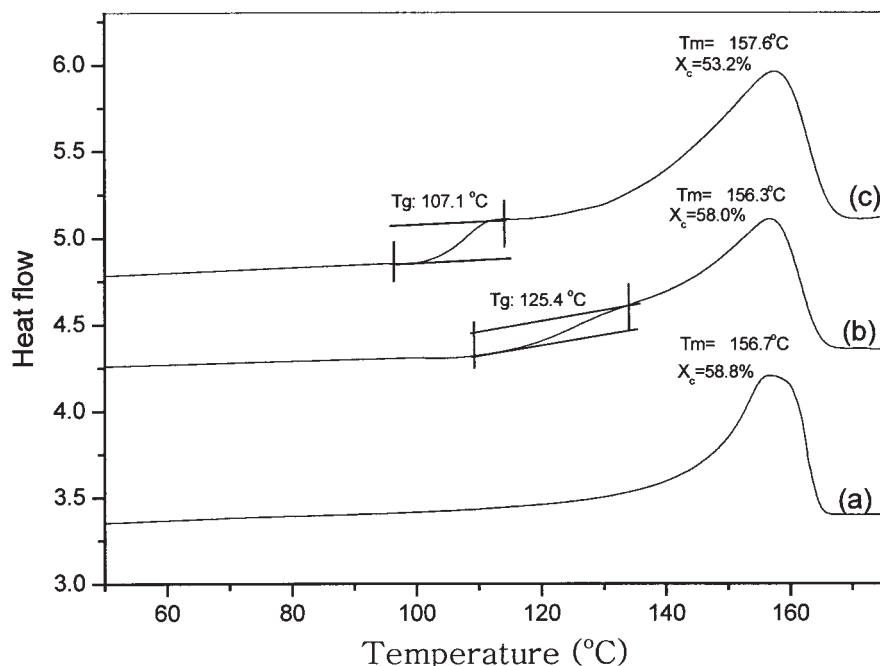


Figure 3. DSC curves of (a) the starting poly(propylene-co-1,4-divinylbenzene) copolymer, (b) the graft copolymer *i*-PP-*g*-PMMA, and (c) the graft copolymer *i*-PP-*g*-PS.

ene-co-1,4-divinylbenzene) copolymer and was 0.53 mol % on average. With the defined graft density, a graft length (M_n) of 24,400 g/mol was determined for the PMMA side chains with the following equation:

$$M_n(\text{g/mol}) = \frac{\text{mass}_{\text{graft copolymer}} - \text{mass}_{\text{starting poly(propylene-co-1,4-divinylbenzene)}}}{\text{moles of pendant vinylbenzene groups}} \quad (1)$$

As for the *i*-PP-*g*-PS graft copolymer, because the initiation efficiency of the graft polymerization was calculated to be 82%, the graft density and graft length of the *i*-PP-*g*-PS graft copolymer were estimated to be 0.44 mol % and 41,220 g/mol, respectively.

Despite the high contents of the second polymeric component, the *i*-PP-*g*-PMMA and *i*-PP-*g*-PS graft copolymers still possessed high T_m 's because of their relatively low graft densities. Figure 3 compares the DSC curves of the *i*-PP-*g*-PMMA and *i*-PP-*g*-PS graft copolymers with that of the original poly(propylene-co-1,4-divinylbenzene). T_g 's of PMMA and PS segments can be observed at 125 and 107 °C, respectively. The PP segments in both graft copolymers show T_m and χ_c values comparable to those of the

starting PP copolymer, and this ensures efficient cocrystallization with bulk *i*-PP when the graft copolymers are used as compatibilizers in *i*-PP-based blends.

CONCLUSIONS

We have demonstrated a convenient yet highly efficient method for the preparation of *i*-PP-based graft copolymers via ATRP. Through the addition of a small amount of vinylbenzene groups into PP chains via the selective copolymerization of propylene with 1,4-divinylbenzene catalyzed by an isospecific Ziegler–Natta catalyst, the multifunctional ATRP initiator of 1-chloroethylbenzene-pending *i*-PP was readily achieved by a quantitative hydrochlorination reaction and was shown to be very effective for the initiation of the graft polymerization of MMA and styrene for the preparation of structurally well-defined *i*-PP-based graft copolymers.

This work was financially supported by the Chinese Academy of Sciences through the One Hundred Talents program and by the National Science Foundation of China (20304017, 50373048, and 20334030).

REFERENCES AND NOTES

- Galli, P.; Vecellio, G. *J Polym Sci Part A: Polym Chem* 2004, 3, 396.
- Corradini, P. *J Polym Sci Part A: Polym Chem* 2004, 3, 391.
- Kashiwa, N. *J Polym Sci Part A: Polym Chem* 2004, 1, 1.
- Kaminsky, W. *J Polym Sci Part A: Polym Chem* 2004, 16, 3911.
- Chung, T. C. *Prog Polym Sci* 2002, 27, 39 and references therein.
- Wang, J. S.; Matyjaszewski, K. *J Am Chem Soc* 1995, 117, 5614.
- Wang, J. S.; Matyjaszewski, K. *Macromolecules* 1995, 28, 7901.
- Matyjaszewski, K.; Xia, J. *Chem Rev* 2001, 101, 2921.
- Kamigaito, M.; Ando, T.; Sawamoto, M. *Chem Rev* 2001, 101, 3689.
- Patten, T. E.; Matyjaszewski, K. *Acc Chem Res* 1999, 32, 895.
- Bicak, N.; Ozlem, M. *J Polym Sci Part A: Polym Chem* 2003, 41, 3457.
- Muehlebach, A.; Rime, F. *J Polym Sci Part A: Polym Chem* 2003, 41, 3425.
- Tian, Y.; Li, Y.; Iyoda, T. *J Polym Sci Part A: Polym Chem* 2003, 41, 1871.
- Lu, J.; Liang, H.; Zhang, W.; Cheng, Q. *J Polym Sci Part A: Polym Chem* 2003, 41, 1237.
- Cokun, M. *J Polym Sci Part A: Polym Chem* 2003, 41, 668.
- Lutz, J. F.; Jahed, N.; Matyjaszewski, K. *J Polym Sci Part A: Polym Chem* 2004, 42, 1939.
- Kazmierski, K.; Hurduc, N.; Sauvet, G.; Chojnowski, J. *J Polym Sci Part A: Polym Chem* 2004, 42, 1682.
- Inoue, Y.; Matyjaszewski, K. *J Polym Sci Part A: Polym Chem* 2004, 42, 496.
- Matyjaszewski, K.; Teodorescu, M.; Miller, P. J.; Peterson, M. L. *J Polym Sci Part A: Polym Chem* 2000, 38, 2440.
- Liu, S.; Sen, A. *Macromolecules* 2001, 34, 1529.
- Hong, S. C.; Jia, S.; Teodorescu, M.; Kowalewski, T.; Matyjaszewski, K.; Gottfried, A. C.; Brookhart, M. *J Polym Sci Part A: Polym Chem* 2002, 40, 2736.
- Yamamoto, K.; Tanaka, H.; Sakaguchi, M.; Shimada, S. *Polymer* 2003, 44, 7661.
- Schulze, U.; Fonagy, T.; Komber, H.; Pompe, G.; Pionteck, J.; Ivan, B. *Macromolecules* 2003, 36, 4719.
- Desai, S. M.; Solanky, S. S.; Mandale, A. B.; Rathore, K.; Singh, R. P. *Polymer* 2003, 44, 7645.
- Fonagy, T.; Ivan, B.; Szesztay, M. *Macromol Rapid Commun* 1998, 19, 479.
- Hong, S. C.; Pakula, T.; Matyjaszewski, K. *Macromol Chem Phys* 2001, 202, 3392.
- Hong, S. C.; Jia, S.; Teodorescu, M.; Kowalewski, T.; Matyjaszewski, K.; Gottfried, A. C.; Brookhart, M. *J Polym Sci Part A: Polym Chem* 2002, 40, 2736.
- Fonagy, T.; Ivan, B.; Szesztay, M. *J Reinf Plast Comp* 2002, 21, 1411.
- Chung, T. C.; Dong, J. Y. *Macromolecules* 2002, 35, 2868.
- The viscosity-average molecular weight (M_η) was calculated as follows: $[\eta] = 1.10 \times 10^{-4} M_\eta^{0.8}$. See Kinsinger, J. B.; Huges, R. E. *J Phys Chem* 1959, 63, 2002.